2025 IEEE/CIC International Conference on Communications in China (ICCC Workshops) | 978-1-6654-7801-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCCWorkshops67136.2025.11148098

AIGC-Based Radio Map Construction for Channel Estimation in Low-Altitude Economy

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***Abstract*—Low-altitude economy (LAE) has garnered increasing interest, but its large-scale deployment faces substantial obstacles. The secure and efficient operations of unmanned aerial vehicles (UAVs) rely on reliable and high-speed wireless communication services delivered by base stations (BSs). However, high mobility, in addition to multipath effects, poses significant challenges to channel estimation. To address these challenges, we propose an AI-generated content (AIGC)-based approach to assist channel estimation for LAE. Specifically, a new generative adversarial network (GAN)-based model named continuous vector-conditioned GAN (CVCGAN) is designed to construct the radio map by establishing the mapping from UAV's sensing information to its channel space. Subsequently, a neural network-based integrator is developed to perform Bayesian inference of channel state information (CSI), leveraging the prior distribution learned by the constructed radio map and the partial observations provided by pilot measurements. Simulation results validate the superiority of the proposed scheme, which outperforms state-of-the-art (SOTA) approaches including ChannelNet, conditional GAN (CGAN), and long short-term memory (LSTM).**

***Index Terms*—AIGC, Channel estimation, GAN, LAE, radio map, UAV.**

I. Introduction

2024 is widely regarded as the inaugural year of the low-altitude economy (LAE). On the one hand, there has been a surging demand for aerial services such as logistics, surveillance, and smart city development recently. On the other hand, advancements in technology particularly in unmanned aerial vehicles (UAVs), wireless communications and sensing, and artificial intelligence (AI), have further propelled the concept of LAE. As a new economic form, LAE refers to various flying activities creating commercial and social values conducted at altitudes below 1,000 meters [1].

Despite the growing interest in LAE, its large-scale deployment remains challenging. Specifically, Safe and efficient UAV operations critically depend on reliable and high-speed wireless communication services delivered by base stations (BSs) [2]. To ensure effective beamforming and interference mitigation, BSs require accurate downlink channel state information (CSI) of UAVs [3]. However, accurate channel estimation for UAVs is challenging due to the following aspects. The high mobility of UAVs introduces Doppler shifts, resulting in a rapidly time-varying channel. In addition, high-rise buildings contribute to multipath effects in UAV communications, causing frequency-selective fading. Furthermore, the interaction of multipath effects and Doppler shifts induces Doppler spread, leading to time-selective fading. The fading that is both frequency- and time-selective is also known as doubly selective fading [4], which increases pilot overhead across both time and frequency domains. Moreover, the application of multiple-input multiple-output (MIMO) further increases the pilot overhead in the spatial domain. The need for pilots in space, time and frequency dimensions makes the pilot overhead unbearable for data transmission.

A. Related Works and Limitations

Traditional channel estimation methods such as least squares (LS), minimum mean-square error (MMSE), and linear MMSE (LMMSE) have been widely adopted in modern wireless communications. These classical methods are simple to implement, but their performance heavily depends on the number of pilots. Compressed sensing (CS) [5] can achieve higher estimation accuracy with limited pilots by utilizing the sparsity of channels. Representative CS-based algorithms include the basis pursuit (BP) [6], subspace pursuit (SP) [7], and orthogonal matching pursuit (OMP) [8]. However, these CS-based algorithms rely on idealized assumptions, e.g., sparsity known a priori and discrete support.

With the rapid advancement of AI and its widespread success in various fields, neural networks have been applied to channel estimation, demonstrating notable performance gains [9]. In [10], a deep neural network (DNN) is employed for joint channel estimation and signal detection, where an end-to-end framework implicitly learns the CSI and directly recovers transmitted symbols. However, the high dimensionality of the wireless channel leads to an excessive number of neural network parameters. In [11], a two-stage pipeline named ChannelNet is proposed, which consists of a super-resolution convolutional neural network (SRCNN) and a denoising CNN (DnCNN). The core idea is to treat the coarse CSI estimated by pilots as a low-resolution image and reconstruct a high-resolution version via CNNs, thereby enhancing estimation accuracy. In [12], a hybrid architecture that combines DNNs and long short-term memory (LSTM) is developed for channel estimation and tracking in UAV communications.

Recently, AI-generated content (AIGC) has emerged as a promising paradigm for wireless communications. In particular, generative models such as generative adversarial networks (GANs) [13] have been applied to channel estimation. In [14], GAN is utilized to learn a mapping from a low-dimensional

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latent space to the high-dimensional channel space. In [15], conditional GANs (CGANs) [16] have been introduced by incorporating received signals and pilot information as conditional inputs. In [17], channel parameters are generated conditioned on the position and velocity information of the UAV, which are then utilized to obtain the channel impulse response (CIR). However, applying CGAN to generate low-dimensional channel parameters may underexploit its generative capacity.

B. Contributions and Paper Organization

In the scenarios of LAE, fixed areas (i.e., air corridors) are assigned for UAVs accounting for safety and management reasons. Besides, UAVs are required to follow the planned routes, with most segments being straight considering energy constraints. These features make it possible to build radio maps [18] for LAE scenarios. Radio map is a promising technology that connects the UE's location with its CSI [19], [20]. In this paper, we introduce velocity information into the radio map and extend its application from static scenarios to high mobility scenarios. In summary, this paper makes the following contributions.

- We develop a new GAN-based model named continuous vector-conditioned GAN (CVCGAN)[1] to construct the radio map. The generator is trained by competing with the discriminator and a pre-trained estimator to produce realistic CSI with given continuous vector labels.
- We design an integrator to fuse the prior information obtained from the generator with the CSI estimated by pilots to perform radio map-assisted channel estimation.
- We validate the performance of the proposed method through numerical simulations. The results demonstrate that the designed CVCGAN can effectively generate realistic channels with given labels. The integrator significantly improves channel estimation performance, outperforming state-of-the-art (SOTA) approaches including ChannelNet [11], CGAN [15] and LSTM [12].

The remainder of the paper is organized as follows. In Section II, the system model of channel estimation in LAE scenarios is presented. In Section III, CVCGAN is proposed to construct the radio map with the collected CSI data labeled with sensing information. In Section IV, an integrator is designed to estimate the CSI with a priori information provided by radio map and partial observations obtained from pilots. In Section V, numerical simulations and results are given, followed by conclusions in Section VI.

Notations: a, $\mathbf{a}$, and $\mathbf{A}$ represent a scalar, a vector, and a matrix, respectively. $(\cdot)^*$ and $(\cdot)^T$, $(\cdot)^H$ represent conjugate, transpose, and conjugate transpose operations, respectively. $\mathcal{N}(\mu, \sigma^2)$ represents Gaussian distribution with the mean μ and the standard deviation σ. $\otimes$ denotes the Kronecker product. $||\cdot||$ denotes the l_2 (Euclidean) norm. ∇ represents the gradient operator. $\mathbb{R}$ and $\mathbb{C}$ stand for the set of real and complex numbers, respectively.

[1]We name our model CVCGAN to distinguish it from CCGAN [21], which does not support conditioning on vector labels as our model does.

II. System Model

In the scenario of LAE, a BS is responsible for providing wireless communication services for UAVs engaged in various applications in the assigned areas.

A. Motion Model

For simplicity, a two-dimensional plane is considered in our model. In addition, UAVs are assumed to follow straight routes for energy issues, and the routes are parallel to each other to avoid possible collisions. A two-dimensional Cartesian coordinate system is established for the airspace, where the x-axis is aligned with the UAV flight path. The initial location of the UAV is given by (x_0, y_0). The UAV's speed varies over time, with the speed at time t denoted as $v(t)$. Therefore, the location of the UAV at time t can be written as

$$\begin{cases} x_t = x_0 + \int_0^t v(\tau)\, d\tau + n_x \\ y_t = y_0 + n_y \end{cases}, \tag{1}$$

where $n_x, n_y \sim \mathcal{N}(0, \sigma^2)$ are additive Gaussian noise terms accounting for course deviations caused by environmental factors such as wind disturbance.

B. Channel Model

We consider a multi-input single-output (MISO)-orthogonal frequency division multiplexing (OFDM) communication system in the model. The BS is equipped with a uniform planar array (UPA) consisting of $N_t = N_{t,x} \times N_{t,y}$ transmit antennas. UAVs are assumed to be single-antenna due to size, weight, and power (SWaP) constraints. The system bandwidth is denoted by B, and the number of subcarriers is K.

UAV communications are impacted by multipath effects caused by urban high-rise buildings as well as the ground. In addition, each propagation path is also affected by Doppler shifts induced by the UAV's motion. As a result, the channel exhibits doubly selective fading in both time and frequency domains, which can be modeled by the delay-Doppler response [22] as

$$\begin{aligned} \boldsymbol{h}(\tau, \nu) = \sum_{l=1}^{L} \sqrt{p_l} \cdot \mathbf{a}_r(\theta_l^r, \phi_l^r) \cdot \mathbf{a}_t^T(\theta_l^t, \phi_l^t) \cdot e^{j\varphi_l} \\ \delta(\tau - \tau_l) \cdot \delta(\nu - \nu_l), \end{aligned} \tag{2}$$

where L is the total number of paths, p_l, φ_l, τ_l, and ν_l are the power, phase, delay, and Doppler of the l-th path, respectively, $\delta(\cdot)$ is the Dirac delta function. $\mathbf{a}_r(\theta_l^r, \phi_l^r)$ and $\mathbf{a}_t(\theta_l^t, \phi_l^t)$ are the receive and transmit array responses, respectively. The parameters θ_l^r,ϕ_l^r,θ_l^t, and ϕ_l^t are the elevation angle of arrival (EAoA), azimuth angle of arrival (AAoA), elevation angle of departure (EAoD), and azimuth angle of departure (AAoD), respectively. In the MISO model, $\mathbf{a}_r = 1$ for single receive antenna, and $\mathbf{a}_t = \mathbf{a}_{t,x} \otimes \mathbf{a}_{t,y}$ for UPA transmit antennas, where $\mathbf{a}_{t,x}$ and $\mathbf{a}_{t,y}$ are the elemental array response vectors along the x-axis and y-axis. The Doppler shift can be calculated by $\nu_l = f_c \frac{v_l}{c}$, where $v_l = v \sin\theta_l \cos\phi_l$ is the radial velocity on the l-th path and c is the speed of light.

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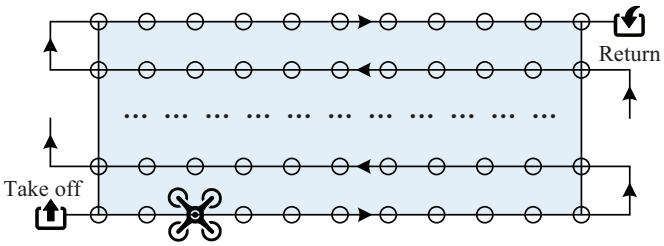

Fig. 1: The UAV channel measurement strategy.

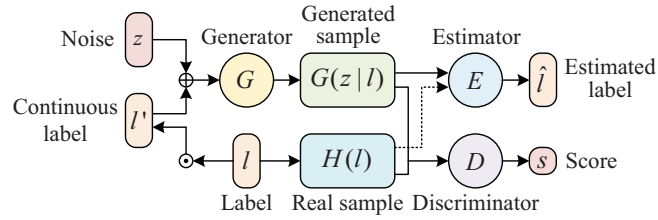

Fig. 2: The architecture of the proposed CVCGAN.

The delay-Doppler response of (2) can be transformed to its time-variant impulse response as

$$\begin{aligned}\boldsymbol{g}(\tau,t) &= \int_{-\infty}^{\infty} \boldsymbol{h}(\tau,\nu)\cdot e^{j2\pi\nu(t-\tau)}\mathrm{d}\nu \\ &= \sum_{l=1}^{L}\sqrt{p_l}\cdot \mathbf{a}_r(\theta_l^r,\phi_l^r)\cdot \mathbf{a}_t^T(\theta_l^t,\phi_l^t)\cdot e^{j\varphi_l} \\ &\quad e^{j2\pi\nu_l(t-\tau)}\cdot\delta(\tau-\tau_l).\end{aligned} \tag{3}$$

Then, the time-frequency response can be obtained by performing the Fourier transform on (3), i.e.,

$$\begin{aligned}\boldsymbol{H}(f,t) &= \int_{-\infty}^{\infty} \boldsymbol{g}(\tau,t)\cdot e^{-j2\pi f\tau}\mathrm{d}\tau \\ &= \sum_{l=1}^{L}\sqrt{p_l}\cdot \mathbf{a}_r(\theta_l^r,\phi_l^r)\cdot \mathbf{a}_t^T(\theta_l^t,\phi_l^t)\cdot e^{j\varphi_l} \\ &\quad e^{-j2\pi\tau_l f}\cdot e^{j2\pi\nu_l t}\cdot e^{-j2\pi\nu_l\tau_l}.\end{aligned} \tag{4}$$

To facilitate the implementation of OFDM systems, the continuous time-frequency channel response is discretized along both the frequency and time domains. In this context, it is assumed that the channel remains approximately constant over the duration of a single subcarrier and one OFDM symbol, i.e., within each time-frequency resolution element. Under this assumption, the discretized channel response can be formulated by

$$\begin{aligned}\mathbf{H}(k\Delta f,mT) &= \sum_{l=1}^{L}\sqrt{p_l}\cdot \mathbf{a}_r(\theta_l^r,\phi_l^r)\cdot \mathbf{a}_t^T(\theta_l^t,\phi_l^t)\cdot e^{j\varphi_l} \\ &\quad e^{-j2\pi\tau_l k\Delta f}\cdot e^{j2\pi\nu_l mT}\cdot e^{-j2\pi\nu_l\tau_l},\end{aligned} \tag{5}$$

where $\Delta f = B/K$ is the subcarrier spacing, $T = 1/\Delta f$ is the period of the OFDM symbol.

III. CVCGAN-Based Radio Map Construction

The features of LAE scenarios, with fixed airspace and planned routes, provide the possibility for radio map construction. To build the radio map, a corridor scan strategy is applied to collect CSI data annotated with location and velocity labels at discrete measurement points. Then, CVCGAN is developed to construct the map by learning a continuous mapping from sensing information to the channel space.

A. Data Acquisition

As illustrated in Fig. 1, the continuous airspace is divided into discrete measurement points. Considering the division for a $d_x \times d_y$ airspace, which is uniformly divided into multiple small grids with spacing Δd. The measurement points are placed at each grid vertex with $N_x = d_x/\Delta d + 1$ vertices along the x-axis and $N_y = d_y/\Delta d + 1$ vertices along the y-axis. Therefore, a total of $N_x \cdot N_y$ measurement points are set for the airspace. Then, the UAV takes off from outside the measurement area and accelerates to the desired speed. It then traverses all measurement points at the specified speed v by applying a corridor scan strategy. After that, by repeating the above process for N_v velocity levels, a total of $N_l = N_x \times N_y \times N_v$ CSI samples $H(\boldsymbol{l})$ with $\boldsymbol{l} = (x, y, v)$ denoting location and velocity labels can be obtained.

B. The Proposed CVCGAN

CGAN was proposed to address the limitation of classical GAN in controlling the modes of data generation. While CGAN is effective for generating data conditioned on discrete scalar labels, it is inherently unsuitable for continuous vector labels due to the following two aspects. On one hand, CGAN can only generate data for trained labels, and since it is not trained for continuous labels, it cannot perform well with continuous labels. On the other hand, CGAN requires label embedding. This operation necessitates knowing the number of all discrete labels, which is obviously not feasible for infinite continuous labels.

To generate the channel samples for continuous positions and velocities, we develop CVCGAN, whose architecture is shown in Fig. 2. In the proposed CVCGAN, a new neural network named estimator is introduced. The added estimator is utilized to estimate the position and velocity information of the channel, and it is trained based on real channel samples before the adversarial training between the generator and the discriminator. The loss function of the estimator is defined as the weighted MSE between the estimated labels and the actual labels, i.e.,

$$\mathcal{L}_{\mathrm{e}} = \left|\left|\boldsymbol{w}\big(E(H(\boldsymbol{l})) - \boldsymbol{l}\big)\right|\right|^2, \tag{6}$$

where $\boldsymbol{w} = (w_x, w_y, w_v)$ represents the weight vector assigned to different estimated parameters. After that, the weights of the estimator are frozen during the following adversarial training.

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The generator produces channel samples with the concatenation of noise $\boldsymbol{z}$ and continuous label $\boldsymbol{l}'$. The continuous label $\boldsymbol{l}'$ is generated by performing interpolation on discrete label $\boldsymbol{l}$, so that $\boldsymbol{l}'$ follows a uniform distribution within the range of $\boldsymbol{l}$. The input $\boldsymbol{l}'$ helps the generator learn how to generate samples based on continuous labels during training, addressing the first limitation of CGAN. During the training of the discriminator, the set of the parameters of the generator θ_{g} is frozen. The generated samples $G(\boldsymbol{z}|\boldsymbol{l})$, as well as the real samples $H(\boldsymbol{l})$, are fed into the discriminator, which is trained to assign a score to evaluate the quality of the input sample. The discriminator does not need to determine whether the sample matches the label, avoiding the label embedding operation and filling the second deficiency of CGAN. After the discriminator is trained, its parameters set θ_{d} is frozen at the training phase of the generator. The generator's output samples are input to both the estimator and the discriminator. The estimator is responsible for inferring the position and velocity information of the generated samples, while the discriminator evaluates the quality of the generated samples. The discriminator and generator are trained iteratively until θ_{g} converges.

To avoid gradient vanishing in classical GANs, Wasserstein GAN (WGAN) [23] with gradient penalty (WGAN-GP) [24] is adopted in the proposed CVCGAN. WGAN applies Wasserstein distance, a.k.a. Earth-Mover (EM) distance instead of Jensen-Shannon (JS) divergence to quantify the difference between two distributions. Besides, gradient penalty is incorporated to enforce the Lipschitz constraint on the discriminator to improve the training stability. The corresponding minimax problem about $V(G,D)$ for the proposed CVCGAN is formulated as

$$\begin{aligned}&\min_{G}\max_{D} V(D,G) = \\ &\quad \underbrace{\mathbb{E}_{H\sim p_{\text{data}}(H)}\big[D(H(\boldsymbol{l}))\big] - \mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}\big[D(G(\boldsymbol{z}|\,\boldsymbol{l}'))\big]}_{\text{Adversarial loss}} - \\ &\quad \underbrace{\lambda_{\text{gp}}\mathbb{E}_{\widetilde{\mathbf{H}}\sim P_{\widetilde{\mathbf{H}}}}\big[\big(\big|\big|\nabla_{\widetilde{\mathbf{H}}}D(\widetilde{\mathbf{H}})\big|\big|-1\big)^2\big]}_{\text{Gradient penalty}} + \\ &\quad \underbrace{\lambda_{\text{e}}\mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}\big|\big|\boldsymbol{w}\big(E(G(\boldsymbol{z}|\,\boldsymbol{l}')) - \boldsymbol{l}'\big)\big|\big|^2}_{\text{Estimate loss}},\end{aligned} \tag{7}$$

where λ_{gp} and λ_{e} are coefficients that control the strength of the gradient penalty and the estimate loss terms, respectively, $\widetilde{\mathbf{H}}$ represents the interpolated samples and can be acquired by performing interpolation between real samples and generated samples, i.e.,

$$\widetilde{\mathbf{H}} = \epsilon\cdot H(\boldsymbol{l}) + (1-\epsilon)\cdot D(G(\boldsymbol{z}|\,\boldsymbol{l}')), \tag{8}$$

where the parameter ϵ is a random number that follows a uniform distribution from 0 to 1. It can be inferred from (7) that the discriminator is required to distinguish real samples from generated samples, assigning high scores to real samples and low scores to generated samples. Simultaneously, the discriminator must maintain its gradient variation. The loss function of the discriminator can be written as

$$\begin{aligned}\mathcal{L}_{\text{d}} &= \mathcal{L}_{\text{da}} + \lambda_{\text{gp}}\cdot\mathcal{L}_{\text{gp}} \\ &= -D(H(\boldsymbol{l})) + D(G(\boldsymbol{z}|\,\boldsymbol{l}')) + \lambda_{\text{gp}}\cdot\big(\big|\big|\nabla_{\widetilde{\mathbf{H}}}D(\widetilde{\mathbf{H}})\big|\big|-1\big)^2,\end{aligned} \tag{9}$$

where $\mathcal{L}_{\text{da}}$ is the adversarial loss of the discriminator. As for the generator, it is required not only to produce realistic samples to deceive the discriminator into awarding a high score, but also to meet the requirements from the estimator, ensuring that the generated samples closely correspond to the provided labels. The loss function of the generator can be expressed as

$$\begin{aligned}\mathcal{L}_{\text{g}} &= \mathcal{L}_{\text{ga}} + \lambda_{\text{e}}\cdot\mathcal{L}'_{\text{e}} \\ &= -D(G(\boldsymbol{z}|\,\boldsymbol{l}')) + \lambda_{\text{e}}\cdot\big|\big|\boldsymbol{w}\big(E(G(\boldsymbol{z}|\,\boldsymbol{l}')) - \boldsymbol{l}'\big)\big|\big|^2,\end{aligned} \tag{10}$$

where $\mathcal{L}_{\text{ga}}$ is the adversarial loss for the generator. Due to page limitations, the detailed architectures of the estimator, discriminator, and generator here, as well as the integrator given in Section IV, are provided in the released code[2].

IV. The Integration of Radio Map and Pilots

After the construction of radio map through the proposed CVCGAN, the channel of the UAV at an arbitrary position and velocity within the built map can be generated, which is denoted by $\mathbf{H}_{\text{m}}$. Due to the high dynamic characteristic of the UAV, the channel exhibits rapidly time-varying properties. Therefore, $\mathbf{H}_{\text{m}}$ obtained from the radio map may significantly deviate from the actual channel $\mathbf{H}$ to be estimated. Although $\mathbf{H}_{\text{m}}$ cannot be directly utilized as an estimate of $\mathbf{H}$, it can serve as prior information regarding the distribution of $\mathbf{H}$. To be specific, channels sampled from the same position at the same speed have the same characteristics. In other words, they can be seen as following an identical distribution, i.e., $H(x,y,v) \sim \mathcal{H}(x,y,v)$, where $\mathcal{H}(x,y,v)$ represents the channel distribution at position (x,y) and velocity v. Therefore, the generated channel $\mathbf{H}_{\text{m}}$ and the actual channel $\mathbf{H}$ follow the same distribution as

$$\mathbf{H}, \mathbf{H}_{\text{m}} \sim \mathcal{H}(x,y,v). \tag{11}$$

In comparison to the radio map, pilot-based measurement offers a real-time approach to channel acquisition. For example, the LS method aims to obtain the pilot-based estimation $\mathbf{H}_{\text{p}}$ by minimizing the squared error between the received pilot signal and its predicted value, i.e.,

$$\mathbf{H}_{\text{p}} = \arg\min_{\mathbf{H}} ||\mathbf{y}_{\text{p}} - \mathbf{H}\mathbf{x}_{\text{p}}||^2, \tag{12}$$

where $\mathbf{x}_{\text{p}}$ and $\mathbf{y}_{\text{p}}$ are pilot symbols and received signal, respectively. Thus, channel estimation based on pilots $\mathbf{H}_{\text{p}}$ can serve as the estimation of $\mathbf{H}$ directly. However, the limited length of the pilot sequence constrains its ability to provide only partial observations of the channel, failing to achieve accurate channel estimation, particularly in scenarios where

[2]https://github.com/yangbin-xd/LAE.git

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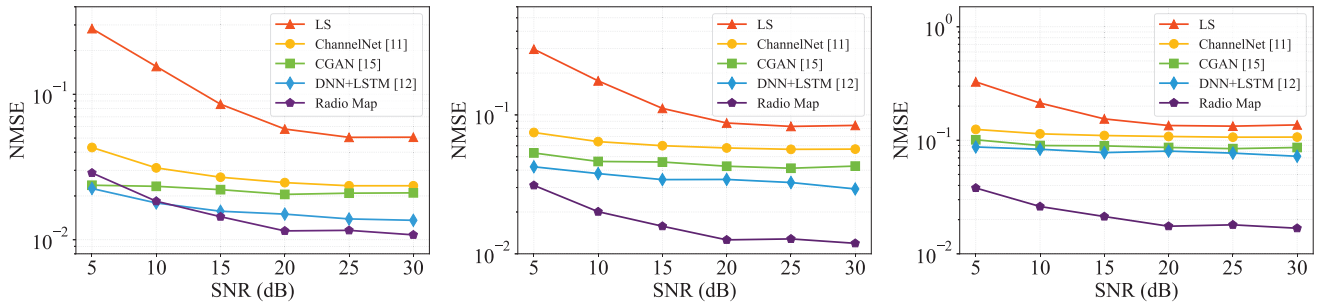

(a) 10m/s (b) 20m/s (c) 30m/s

Fig. 3: The NMSEs of the radio map compared with other approaches versus average SNRs at different velocities.

the channels experience doubly selective fading. To address this issue, the prior distribution information provided by the radio map is integrated with partial observations obtained from pilot measurements. According to Bayes' theorem, a posterior estimation of the channel can be obtained by incorporating prior information with limited observations.

However, the prior distribution of $\mathcal{H}(x, y, v)$ is learned by the radio map and is challenging to represent explicitly. To avoid giving expressions of the distribution, a neural network-based integrator is designed to perform Bayesian estimation implicitly with $\mathbf{H}_{\mathrm{m}}$ and $\mathbf{H}_{\mathrm{p}}$ as input, i.e.,

$$\hat{\mathbf{H}} = I\left(\mathbf{H}_{\mathrm{m}}, \mathbf{H}_{\mathrm{p}}\right). \tag{13}$$

The objective of the integrator is to estimate $\mathbf{H}$ accurately, so the loss function is customized to be the MSE between the network output and the target channel, i.e.,

$$\mathcal{L}_{\mathrm{i}} = \left|\left|I(\mathbf{H}_{\mathrm{m}}, \mathbf{H}_{\mathrm{p}}) - \mathbf{H}\right|\right|^2. \tag{14}$$

V. Numerical Simulations and Results

In this section, the detailed simulation settings are provided, followed by the presentation and discussion of the results.

A. Dataset and Parameters

In our setup, an airspace region of 200×100m is uniformly partitioned with an interval of $\Delta d = 10$m. Consequently, $N_x = 21$ vertices along the x-axis and $N_y = 11$ vertices along the y-axis can be obtained, yielding a total of $N_x \cdot N_y = 231$ measurement points. After that, a corridor scan strategy is employed by the UAV to conduct channel measurement at each point. The speeds of the UAV for measurement are set to six levels containing 5, 10, 15, 20, 25 and 30m/s. The UAV acquires CSI annotated with geographical position and radial velocity labels by traversing different measurement points with varying flight speeds.

The communication parameters are set as follows. The carrier frequency f_c is 3.5GHz, which serves as a primary frequency for 5G. The configuration of the subcarrier spacing and the number of subcarriers follows the specifications in 3GPP TS 38.104 of release 18. The subcarrier spacing is set to be 15kHz, and the number of subcarriers is 624, corresponding to 52 resource blocks (RBs). For simplicity, the first RB, which consists of $N_c = 12$ consecutive subcarriers, is focused on. During the measurement, an OFDM frame is collected at each measurement point, which contains $N_f = 10$ OFDM subframes, and each subframe has $N_s = 14$ OFDM symbols. The average SNR of the CSI collected at all measurement points is assumed to be 30dB, rather than being considered noise-free. A UPA is deployed at the BS with $8 \times 4 = 32$ transmit antennas, while the UAV is equipped with a single receive antenna. The pilot length is 32, which is equal to the number of transmit antennas. The CSI used in this paper is synthesized through a ray-tracing simulator[3].

B. Numerical Results

The performance of channel estimation is evaluated by normalized MSE (NMSE). The NMSEs of the radio map with other approaches under different average SNRs at different velocities are compared in Fig. 3, where LS method serves as the baseline and three benchmarks consist of ChannelNet [11], CGAN [15] and DNN+LSTM [12] are chosen. In Fig. 3(a) where the velocity is 10m/s, CGAN and DNN+LSTM demonstrate superior performance under low SNR conditions. However, when the average SNR exceeds 10dB, the proposed radio map approach begins to outperform these benchmarks. Specifically, in Fig. 3(b), with the UAV flying at 20m/s, the radio map approach consistently outperforms all other methods across the average SNR range from 5 to 30dB, and the performance gap widens as the SNR increases. In Fig. 3(c), where the speed reaches 30m/s, the NMSEs of the benchmarks remain around 0.1, getting limited profit from increasing SNR. This is due to the rapid variation of the channel, which makes it difficult to accurately estimate the CSI using only pilot-based partial observations. In contrast, the radio map approach leverages sensing information to effectively learn the prior distribution of the channel. As a result, its estimation performance improves significantly with increasing SNR, reaching nearly an order of magnitude better than these benchmarks.

In addition, we also compare the performances of these approaches versus velocities at different average SNRs in Fig. 4. It can be observed from Fig. 4 that the performances of all

[3]https://github.com/yangbin-xd/ray-tracing.git

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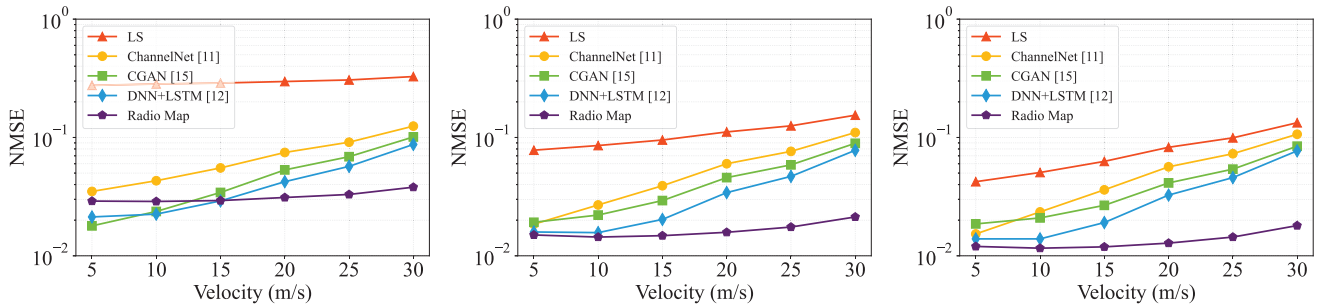

(a) 5dB (b) 15dB (c) 25dB

Fig. 4: The NMSEs of the radio map compared with other approaches versus velocities at different average SNRs.

methods deteriorate as velocity increases. Fig. 4(a) presents the results at an average SNR of 5dB. At this noise level, the classical LS method suffers from severe performance degradation. CGAN and DNN+LSTM methods outperform others when the UAV speed is below 15m/s. However, when the speed exceeds 15m/s, the proposed radio map approach begins to show superior performance. As the average SNR increases, the advantage of the radio map approach becomes increasingly dominant. To be specific, in Figs. 4(b) and 4(c), the radio map method consistently outperforms three benchmarks across all velocity levels. In addition, radio map demonstrates greater robustness to increasing velocities, maintaining stable performance even in high-mobility scenarios. This suggests that radio map effectively learns prior information and translates it into improved channel estimation performance.

VI. Conclusion

In this paper, we propose a paradigm that utilizes radio map to assist channel estimation for LAE. First, an improved GAN-based model named CVCGAN is proposed to construct the radio map by building the connection between sensing information and channel distribution. After that, a neural network-based integrator is developed to perform channel estimation with the prior information provided by the radio map and partial observations obtained from pilot measurements. The proposed method outperforms SOTA approaches including ChannelNet, CGAN, and LSTM, offering a promising solution for channel estimation in LAE scenarios.

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